HILBERT 90 AND KUMMER THEORY, WITH SOLUTIONS

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Hilbert's Theorem 90 says that $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$. Applied to the Kummer sequence $1 \to \mu_m \to \mathbb{G}_{\mathrm{m}} \xrightarrow{m} \mathbb{G}_{\mathrm{m}} \to 1$, it gives an isomorphism $k^\times/k^{\times m} \cong H^1(k, \mu_m)$ with an explicit formula, and, when k contains the m-th roots of unity, a classification of the abelian extensions of k of exponent dividing m. The problems below prove all of this from the definitions in [Sil09, Appendix B]. Problem 1.3 computes cohomology groups by hand, Problem 2.5 proves Hilbert 90, Problem 3.4 unwinds the Kummer sequence and Problem 4.2 derives Kummer theory. Problems 5.4 and 5.7 compute the groups $k^\times/k^{\times m}$ for finite, local and global fields, and cut out a finite subgroup by conditions at the primes. Later, the same sequence for an elliptic curve E will give an injection $E(k)/m \hookrightarrow H^1(k, E[m])$. Then there is no Hilbert 90, and its failure is measured by $H^1(k, E)$. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A field k, not assumed perfect, a separable closure k^{sep} of k, and the absolute Galois group $\mathrm{Gal}_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$. For an extension L of k inside k^{sep}, $\mathrm{Gal}_L := \mathrm{Gal}(k^{\mathrm{sep}}/L)$. (Silverman's Appendix B assumes k perfect and works with an algebraic closure. Nothing in it changes with k^{sep} in its place.)
- A profinite group G, often finite, and a discrete G-module A: an abelian group with a left action of G, written $a \mapsto ga$, such that $G \times A \to A$ is continuous for the discrete topology on A. (Silverman writes right actions, $a \mapsto a^\sigma$.)
- Elements g, h of G, and elements σ, τ, γ of a Galois group such as Gal_k or $\mathrm{Gal}(L/k)$. Where σ is a generator of a cyclic group, it keeps that meaning.
- An integer $m \geqslant 1$ with $\mathrm{char}\, k \nmid m$, and the algebraic group μ_m of m-th roots of unity, the kernel of $x \mapsto x^m$ on $\mathbb{G}_{\mathrm{m}}$. For a field L containing k, $\mu_m(L)$ is the group of m-th roots of unity in L.
- **An abuse of notation.** For a commutative algebraic group A over k, such as $\mathbb{G}_{\mathrm{m}}$, μ_m or an elliptic curve, we write $H^i(k, A) := H^i(\mathrm{Gal}_k, A(k^{\mathrm{sep}}))$, the cohomology of the Gal_k-module of k^{sep}-points of A. For instance,

$$H^1(k, \mathbb{G}_{\mathrm{m}}) = H^1(\mathrm{Gal}_k, (k^{\mathrm{sep}})^\times), \qquad H^1(k, \mu_m) = H^1(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}})).$$

  This is the only place where A stands for $A(k^{\mathrm{sep}})$.
- The quotient $B/m := B/mB$ of an abelian group B. For the multiplicative group of a field L it is $L^\times/L^{\times m}$, where $L^{\times m}$ is the subgroup of m-th powers.

How to use this. Allow about 15 minutes for Problem 1.3, 25 for Problem 2.5 and 25 for Problem 3.4. The heart of Problem 2.5 is items (1) and (4), and that of Problem 3.4 is items (2) and (3). If time is short, leave Problem 2.5(5) and Problem 3.4(5) for later, like every item marked $\star$. Problems 4.2, 5.4 and 5.7 are for home. The `Magma` blocks run in seconds in the free online calculator. Everything else is by hand.

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1. Cocycles by hand

This section uses the definitions of [Sil09, §§B.1–B.2] and [Poo02, §§2–3]. Its last item is [Sil09, Exercise B.2(a)].

Definition 1.1 (Cohomology in degrees 0 and 1 [Sil09, §§B.1–B.2], [Poo02, §3])**.** Let A be a discrete G-module. Then $H^0(G,A) \coloneqq A^G$, the subgroup of the $a \in A$ with $ga = a$ for every $g \in G$. A left 1-cocycle, or simply 1-cocycle, is a continuous map $\xi\colon G \to A$, $g \mapsto \xi_g$, such that

$$\xi_{gh} = \xi_g + g\xi_h \qquad \text{for all } g,h \in G.$$

A left 1-coboundary, or simply 1-coboundary, is a map of the form $g \mapsto ga - a$, for some $a \in A$. Both are *left*: they are written for the left action $a \mapsto ga$. Silverman's are the right ones, for $a \mapsto a^\sigma$: $\xi_{\sigma\tau} = \xi_\sigma^\tau + \xi_\tau$ and $\xi_\sigma = a^\sigma - a$. The 1-cocycles form a group $Z^1(G,A)$ under addition of values, the 1-coboundaries a subgroup $B^1(G,A)$, and $H^1(G,A) \coloneqq Z^1(G,A)/B^1(G,A)$. If G is finite, every map $G \to A$ is continuous. When the group law of A is written multiplicatively, as for $L^\times$ or $\mu_m(L)$, the conditions read $\xi_{gh} = \xi_g \cdot g(\xi_h)$ and $\xi_g = g(a)/a$.

Proposition 1.2 (Trivial action [Sil09, Remarks B.1.1 and B.2.2], [Poo02, §3])**.** *If G acts trivially on A, then $H^0(G,A) = A$ and $H^1(G,A) = \mathrm{Hom}_{\mathrm{conts}}(G,A)$, the group of continuous homomorphisms.*

Problem 1.3 (Computing H^1 from the definition)**.**
In this problem G is finite.

(1) Let $G = \{1,\sigma\}$ have order 2, acting on $\mathbb{Z}$ by $\sigma a = -a$. Compute $H^0(G,\mathbb{Z})$ and $H^1(G,\mathbb{Z})$. Compare with the trivial action of G on $\mathbb{Z}$.

Solution. **$H^0 = 0$ and $H^1 \cong \mathbb{Z}/2\mathbb{Z}$. For the trivial action, $H^0 = \mathbb{Z}$ and $H^1 = 0$.** Here $\mathbb{Z}^G = \{a : -a = a\} = 0$. Every 1-cocycle has $\xi_1 = 0$, since $\xi_1 = \xi_{1\cdot 1} = \xi_1 + \xi_1$. So it is determined by $x \coloneqq \xi_\sigma$, and the only other condition, from $\sigma^2 = 1$, is $0 = \xi_1 = \xi_\sigma + \sigma\xi_\sigma = x - x$, which always holds. So $Z^1(G,\mathbb{Z}) \cong \mathbb{Z}$, via $\xi \mapsto \xi_\sigma$. The coboundary of a has $\xi_\sigma = \sigma a - a = -2a$, so $B^1(G,\mathbb{Z}) \cong 2\mathbb{Z}$ and $H^1(G,\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$. For the trivial action, Proposition 1.2 gives $H^0 = \mathbb{Z}$ and $H^1 = \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z},\mathbb{Z}) = 0$, as $\mathbb{Z}$ has no element of order 2.

(2) Let $G = \langle\sigma\rangle$ be cyclic of order n, and $N \coloneqq 1 + \sigma + \cdots + \sigma^{n-1}$, acting on A. Show that $\xi \mapsto \xi_\sigma$ induces an isomorphism

$$H^1(G,A) \cong \ker(N\colon A \to A)/(\sigma - 1)A.$$

Solution. Let ξ be a 1-cocycle and $x \coloneqq \xi_\sigma$. By induction on i, using $\xi_{\sigma^{i+1}} = \xi_\sigma + \sigma\xi_{\sigma^i}$,

$$\xi_{\sigma^i} = (1 + \sigma + \cdots + \sigma^{i-1})x \qquad (i \geqslant 1). \tag{1.1}$$

So ξ is determined by x, and $Nx = \xi_{\sigma^n} = \xi_1 = 0$. Conversely, let $x \in \ker N$, and define ξ_{σ^i} by (1.1) for every $i \geqslant 0$ (the empty sum for $i = 0$). Replacing i by $i + n$ adds $\sigma^i Nx = 0$, so ξ is well defined on G. It is a 1-cocycle, because

$$\xi_{\sigma^{i+j}} = (1 + \cdots + \sigma^{i-1})x + \sigma^i(1 + \cdots + \sigma^{j-1})x = \xi_{\sigma^i} + \sigma^i\xi_{\sigma^j}.$$

So $\xi \mapsto \xi_\sigma$ is an isomorphism $Z^1(G,A) \to \ker N$. It maps the coboundary of a to $\sigma a - a$, so $B^1(G,A)$ maps onto $(\sigma - 1)A$.

(3) Let $G = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1,c\}$. Use (2) to compute $H^1(G,\mathbb{C}^\times)$ and $H^1(G,\mu_m(\mathbb{C}))$. Compare the order of the second group with that of $\mathbb{R}^\times/\mathbb{R}^{\times m}$.

Solution. **$H^1(G,\mathbb{C}^\times) = 1$, and $\#H^1(G,\mu_m(\mathbb{C})) = \gcd(2,m) = \#\mathbb{R}^\times/\mathbb{R}^{\times m}$.** In multiplicative notation, $Nz = z\bar{z} = |z|^2$ and $(c-1)A$ is the set of quotients $\bar{z}/z$. For $A = \mathbb{C}^\times$,

$\ker N$ is the unit circle. Every w on it is a quotient: if $w \neq -1$, then $z := 1 + \bar{w}$ is nonzero and $\bar{z}/z = (1+w)/(1+w^{-1}) = w$, and $-1 = \bar{\imath}/i$. So $H^1(G, \mathbb{C}^\times) = 1$.
For $A = \mu_m(\mathbb{C})$, complex conjugation is inversion, so $N\zeta = \zeta\zeta^{-1} = 1$ and $\ker N = \mu_m(\mathbb{C})$, while $\bar{\zeta}/\zeta = \zeta^{-2}$ and $(c-1)\mu_m(\mathbb{C}) = \mu_m(\mathbb{C})^2$. The cyclic group $\mu_m(\mathbb{C})$ of order m has $\mu_m(\mathbb{C})/\mu_m(\mathbb{C})^2$ of order $\gcd(2, m)$. On the other side, every positive real number is an m-th power, and -1 is an m-th power in $\mathbb{R}$ exactly when m is odd. So $\mathbb{R}^\times/\mathbb{R}^{\times m}$ also has order $\gcd(2, m)$.

Remark. This is no accident: Problem 3.4 shows that $H^1(k, \mu_m) \cong k^\times/k^{\times m}$ for every field k, and the first computation is Hilbert 90 for $\mathbb{C}/\mathbb{R}$.

(4) Suppose that G has order n. Show that $nH^1(G, A) = 0$.

Solution. Let ξ be a 1-cocycle and $a := \sum_{h\in G} \xi_h$. As h runs over G, so does gh, so summing the cocycle condition over h gives $a = n\xi_g + ga$. Hence $n\xi_g = g(-a) - (-a)$, and $n\xi$ is the coboundary of $-a$.

2. Hilbert 90

This section is [Poo17, Proposition 1.3.15(ii) and Exercises 1.10–1.11], with the proof of [Mil20, Proposition IV.1.3] (also [Mil22, Theorem 5.23]). [Sil09, Proposition B.2.5] states the result and refers to Serre for the proof. Item (2) is the form in which Hilbert stated it.

Theorem 2.1 (Dedekind [Mil22, Theorem 5.14])**.** *Let F be a field and H a group. If $\chi_1, \dots, \chi_r$ are distinct homomorphisms $H \to F^\times$, then they are linearly independent over F: if $a_1\chi_1 + \cdots + a_r\chi_r = 0$ as a function $H \to F$, with $a_i \in F$, then $a_1 = \cdots = a_r = 0$.*

Definition 2.2 (Norm and trace)**.** Let L/k be a finite Galois extension with group G. The norm and the trace of $a \in L$ are $N_{L/k}(a) := \prod_{\sigma\in G} \sigma(a)$ and $\mathrm{Tr}_{L/k}(a) := \sum_{\sigma\in G} \sigma(a)$. Both lie in k.

Proposition 2.3 (Infinite Galois theory)**.**

(1) The subgroups Gal_L, for L running over the finite Galois extensions of k inside k^{sep}, are open and normal in Gal_k, and they form a basis of open neighborhoods of 1 [Sil09, §B.2].
(2) The maps $H \mapsto (k^{\mathrm{sep}})^H$ and $L \mapsto \mathrm{Gal}_L$ are inverse bijections between the closed subgroups of Gal_k and the extensions of k inside k^{sep}. The subgroup H is open if and only if $(k^{\mathrm{sep}})^H$ is finite over k, and then $[\mathrm{Gal}_k : H] = [(k^{\mathrm{sep}})^H : k]$. It is normal if and only if $(k^{\mathrm{sep}})^H$ is Galois over k, and then $\mathrm{Gal}((k^{\mathrm{sep}})^H/k) \cong \mathrm{Gal}_k/H$ by restriction. In particular, $(k^{\mathrm{sep}})^{\mathrm{Gal}_k} = k$ [Mil22, Theorem 7.13].

Definition 2.4 (Restriction and inflation [Sil09, §B.2], [Poo02, §4])**.** Let A be a discrete G-module and $H \subseteq G$ a closed subgroup. The restriction $\mathrm{Res}\colon H^1(G, A) \to H^1(H, A)$ maps the class of ξ to the class of $\xi|_H$. If H is normal, the inflation $\mathrm{Inf}\colon H^1(G/H, A^H) \to H^1(G, A)$ maps the class of ξ to the class of the composition $G \to G/H \xrightarrow{\xi} A^H \subseteq A$. A 1-cocycle of G of this form is inflated from G/H.

Problem 2.5 (Hilbert's Theorem 90)**.**

(1) Let L/k be a finite Galois extension with group G, and $\xi\colon G \to L^\times$ a 1-cocycle. Show that there is a $c \in L^\times$ with

$$b := \sum_{\tau\in G} \xi_\tau \cdot \tau(c) \neq 0,$$

and that then $\xi_\sigma = b/\sigma(b)$ for every $\sigma \in G$. Deduce that $H^1(G, L^\times) = 1$.

SOLUTION. The elements of G restrict to distinct homomorphisms $L^\times \to L^\times$, and the coefficients ξ_τ are nonzero. By Theorem 2.1, $\sum_\tau \xi_\tau \cdot \tau$ is not the zero function on $L^\times$, so some $c \in L^\times$ has $b \neq 0$. For $\sigma \in G$, the cocycle condition gives $\sigma(\xi_\tau) = \xi_\sigma^{-1}\xi_{\sigma\tau}$, so

$$\sigma(b) = \sum_{\tau\in G} \sigma(\xi_\tau)\cdot(\sigma\tau)(c) = \xi_\sigma^{-1}\sum_{\tau\in G}\xi_{\sigma\tau}\cdot(\sigma\tau)(c) = \xi_\sigma^{-1}b,$$

since $\sigma\tau$ runs over G with τ. So $\xi_\sigma = b/\sigma(b) = \sigma(b^{-1})/b^{-1}$, the coboundary of b^{-1}.

(2) Suppose moreover that $G = \langle\sigma\rangle$ is cyclic. Show that if $a \in L^\times$ has $N_{L/k}(a) = 1$, then $a = \sigma(b)/b$ for some $b \in L^\times$.

SOLUTION. By Problem 1.3(2), in multiplicative notation, $H^1(G, L^\times)$ is the group of the a with $N_{L/k}(a) = 1$ modulo the quotients $\sigma(b)/b$. It is trivial by (1).

(3) Let (x, y) be a rational point of the circle $x^2 + y^2 = 1$. Apply (2) to $\mathbb{Q}(i)/\mathbb{Q}$ to find $u, v \in \mathbb{Q}$, not both 0, with

$$(x, y) = \Big(\frac{u^2 - v^2}{u^2 + v^2}, \frac{2uv}{u^2 + v^2}\Big).$$

SOLUTION. Let σ be complex conjugation, which generates $\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q})$. The element $a \coloneqq x+iy$ has $N(a) = x^2 + y^2 = 1$, so $a = \sigma(\beta)/\beta$ for some $\beta \in \mathbb{Q}(i)^\times$. Write $\beta = u - iv$ with $u, v \in \mathbb{Q}$ not both 0. Then

$$x + iy = \frac{u + iv}{u - iv} = \frac{(u + iv)^2}{u^2 + v^2} = \frac{u^2 - v^2}{u^2 + v^2} + i\,\frac{2uv}{u^2 + v^2}.$$

(4) Let $\xi\colon \mathrm{Gal}_k \to (k^{\mathrm{sep}})^\times$ be a continuous 1-cocycle.

(a) Show that there is a finite Galois extension L of k inside k^{sep} such that ξ takes values in $L^\times$ and $\xi_{\sigma\tau} = \xi_\sigma$ for all $\sigma \in \mathrm{Gal}_k$ and $\tau \in \mathrm{Gal}_L$. So ξ is inflated from a 1-cocycle $\mathrm{Gal}(L/k) \to L^\times$.

SOLUTION. Since ξ is continuous and $(k^{\mathrm{sep}})^\times$ is discrete, $\xi^{-1}(1)$ is open, and it contains 1, because $\xi_1 = \xi_1 \cdot \xi_1$. By part (1) of Proposition 2.3, it contains Gal_{L_0} for some finite Galois extension L_0 of k. The fibers of ξ are open and cover the compact group Gal_k, so ξ takes finitely many values, and they lie in some finite extension of k inside k^{sep}. Let L be the Galois closure of the compositum of this extension and L_0, a finite Galois extension of k with $\mathrm{Gal}_L \subseteq \mathrm{Gal}_{L_0}$. For $\tau \in \mathrm{Gal}_L$, $\xi_\tau = 1$, so $\xi_{\sigma\tau} = \xi_\sigma \cdot \sigma(\xi_\tau) = \xi_\sigma$. So ξ factors through $\mathrm{Gal}_k / \mathrm{Gal}_L \cong \mathrm{Gal}(L/k)$ (Proposition 2.3), with values in $L^\times = ((k^{\mathrm{sep}})^\times)^{\mathrm{Gal}_L}$, and the cocycle condition for ξ is the one for the induced map $\mathrm{Gal}(L/k) \to L^\times$.

(b) Deduce that $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$.

SOLUTION. By (4)(a) and (1), there is $b \in L^\times$ with $\xi_\sigma = b/\sigma(b)$ for every $\sigma \in \mathrm{Gal}_k$ (the action of σ on L is that of its image in $\mathrm{Gal}(L/k)$). So ξ is the coboundary of b^{-1}.

Remark. The same argument shows that $H^1(\mathrm{Gal}_k, A) = \varinjlim_L H^1(\mathrm{Gal}(L/k), A^{\mathrm{Gal}_L})$ for every discrete Gal_k-module A [Sil09, Exercise B.4].

(5) ($\star$) Let L/k be finite Galois with group G, and $\xi\colon G \to L$ a 1-cocycle for the additive group of L. Show that $H^1(G, L) = 0$, by a variant of (1).

SOLUTION. By Theorem 2.1 with all coefficients 1, $\mathrm{Tr}_{L/k} = \sum_\tau \tau$ is not the zero function, so some $c \in L$ has $t \coloneqq \mathrm{Tr}_{L/k}(c) \neq 0$, and $t \in k$. Let $b \coloneqq t^{-1}\sum_\tau \xi_\tau \cdot \tau(c)$. Then $\sigma(\xi_\tau) = \xi_{\sigma\tau} - \xi_\sigma$, so

$$\sigma(b) = t^{-1}\sum_{\tau\in G}(\xi_{\sigma\tau} - \xi_\sigma)\cdot(\sigma\tau)(c) = b - \xi_\sigma t^{-1}\sum_{\tau\in G}(\sigma\tau)(c) = b - \xi_\sigma.$$

So $\xi_\sigma = \sigma(-b) - (-b)$.

Remark. [Poo17, Proposition 1.3.15(i)] deduces this, and the vanishing of all the higher groups, from the normal basis theorem. As in (4), it gives $H^1(k, \mathbb{G}_a) = 0$.

3. The Kummer sequence

This section is the exercise in [Poo02, §3], with the explicit formula of [Sil09, Propositions B.2.5(c) and VIII.2.2]. Hilbert's Theorem 90, $H^1(k, \mathbb{G}_m) = 1$, is Problem 2.5(4).

Proposition 3.1 (Finite multiplicative groups [Mil22, Exercise 1-3])**.** *Every finite subgroup of the multiplicative group of a field is cyclic.*

Proposition 3.2 (The long exact sequence [Sil09, Propositions B.1.2 and B.2.3], [Poo02, Theorem 1])**.** *Let $0 \to A \to B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete G-modules. There is an exact sequence*

$$0 \to A^G \to B^G \to C^G \xrightarrow{\delta} H^1(G, A) \to H^1(G, B) \to H^1(G, C),$$

where, for $c \in C^G$, $\delta(c)$ is the class of $g \mapsto gb - b$ for any $b \in B$ with $\psi(b) = c$.

Proposition 3.3 (Inflation–restriction [Sil09, Proposition B.2.4])**.** *Let A be a discrete Gal_k-module and L/k a finite Galois extension inside k^{sep}. The sequence*

$$0 \to H^1(\mathrm{Gal}(L/k), A^{\mathrm{Gal}_L}) \xrightarrow{\mathrm{Inf}} H^1(\mathrm{Gal}_k, A) \xrightarrow{\mathrm{Res}} H^1(\mathrm{Gal}_L, A)$$

is exact.

Problem 3.4 (The Kummer sequence)**.**
Recall that $\mathrm{char}\, k \nmid m$, so $m \neq 0$ in k.

(1) Show that $\mu_m(k^{\mathrm{sep}})$ is cyclic of order m, and that

$$1 \to \mu_m(k^{\mathrm{sep}}) \to (k^{\mathrm{sep}})^\times \xrightarrow{x \mapsto x^m} (k^{\mathrm{sep}})^\times \to 1$$

is an exact sequence of discrete Gal_k-modules. Deduce from Proposition 3.2 that its connecting map induces an isomorphism $\delta\colon k^\times/k^{\times m} \to H^1(k, \mu_m)$.

Solution. For $a \in k^\times$, or more generally $a \in (k^{\mathrm{sep}})^\times$, the polynomial $x^m - a$ is separable: its derivative mx^{m-1} vanishes only at 0, since $m \neq 0$ in k, and 0 is not a root. So it has m distinct roots, all in k^{sep}. For $a = 1$ this shows that $\mu_m(k^{\mathrm{sep}})$ has order m. It is cyclic by Proposition 3.1. For general a it shows that $x \mapsto x^m$ is surjective on $(k^{\mathrm{sep}})^\times$. Its kernel is $\mu_m(k^{\mathrm{sep}})$, and it commutes with the action of Gal_k, as $\sigma(x^m) = (\sigma(x))^m$. The long exact sequence reads

$$k^\times \xrightarrow{m} k^\times \xrightarrow{\delta} H^1(k, \mu_m) \to H^1(k, \mathbb{G}_m) = 1,$$

so δ is surjective with kernel $k^{\times m}$.

(2) Let $a \in k^\times$ and $\alpha \in k^{\mathrm{sep}}$ with $\alpha^m = a$. Check directly, without Proposition 3.2, that $\xi_\sigma := \sigma(\alpha)/\alpha$ is a continuous 1-cocycle with values in $\mu_m(k^{\mathrm{sep}})$, whose class does not depend on the choice of α. Show that $a \mapsto [\xi]$ is a homomorphism $k^\times \to H^1(k, \mu_m)$ with kernel $k^{\times m}$. This is the map δ of (1).

Solution. Since $(\sigma(\alpha))^m = \sigma(a) = a$, $\xi_\sigma \in \mu_m(k^{\mathrm{sep}})$. The cocycle condition holds: $\xi_{\sigma\tau} = (\sigma\tau)(\alpha)/\alpha = (\sigma(\alpha)/\alpha) \cdot \sigma(\tau(\alpha)/\alpha) = \xi_\sigma \cdot \sigma(\xi_\tau)$. And $\xi_{\sigma\tau} = \xi_\sigma$ for τ in the open subgroup $\mathrm{Gal}_{k(\alpha)}$, so ξ is locally constant, that is, continuous. The other choices of α are the $\zeta\alpha$ with $\zeta \in \mu_m(k^{\mathrm{sep}})$, and they replace ξ by $\xi \cdot (\sigma \mapsto \sigma(\zeta)/\zeta)$, which differs from it by a coboundary.

If $\beta^m = b$, then $(\alpha\beta)^m = ab$ and $\sigma(\alpha\beta)/(\alpha\beta) = \xi_\sigma \cdot \sigma(\beta)/\beta$, so the map is a homomorphism. Its kernel: if $\sigma(\alpha)/\alpha = \sigma(\zeta)/\zeta$ for some $\zeta \in \mu_m(k^{\mathrm{sep}})$ and every σ, then $\alpha/\zeta \in (k^{\mathrm{sep}})^{\mathrm{Gal}_k} = k$ (Proposition 2.3), and $a = (\alpha/\zeta)^m \in k^{\times m}$. Conversely, if $a = c^m$ with $c \in k^\times$, choose $\alpha = c$, and $\xi = 1$. Finally, α lifts a along $x \mapsto x^m$, so by Proposition 3.2 this is the formula for δ.

(3) Show directly from Hilbert 90 that every continuous 1-cocycle $\xi\colon \mathrm{Gal}_k \to \mu_m(k^{\mathrm{sep}})$ is cohomologous to one of the form of (2).

Solution. View ξ as a 1-cocycle with values in $(k^{\mathrm{sep}})^\times$. By Hilbert 90 it is a coboundary there: $\xi_\sigma = \sigma(\beta)/\beta$ for some $\beta \in (k^{\mathrm{sep}})^\times$. Then $\sigma(\beta^m)/\beta^m = \xi_\sigma^m = 1$ for every σ, so $a := \beta^m \in k^\times$, and ξ is the cocycle of (2) for a and $\alpha = \beta$.

Remark. So only surjectivity needs Hilbert 90. In the analogous sequence for an elliptic curve, $H^1(k, E)$ replaces $H^1(k, \mathbb{G}_{\mathrm{m}})$, and it is not zero in general.

(4) Let $m = 2$, with $\mathrm{char}\, k \neq 2$. Show that $H^1(k, \mu_2) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \{\pm 1\})$, and that for $d \in k^\times - k^{\times 2}$, $\delta(d)$ is the homomorphism $\chi_d\colon \sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$, with kernel $\mathrm{Gal}_{k(\sqrt{d})}$. Deduce that $d \mapsto k(\sqrt{d})$ is a bijection from the nontrivial elements of $k^\times/k^{\times 2}$ to the quadratic extensions of k inside k^{sep}.

Solution. The action of Gal_k on $\mu_2(k^{\mathrm{sep}}) = \{\pm 1\} \subseteq k$ is trivial, so the first claim is Proposition 1.2. The formula for $\delta(d)$ is (2), and $\chi_d(\sigma) = 1$ exactly when σ fixes $\sqrt{d}$, that is, when $\sigma \in \mathrm{Gal}_{k(\sqrt{d})}$. A nontrivial continuous homomorphism $\mathrm{Gal}_k \to \{\pm 1\}$ is determined by its kernel, an open subgroup of index 2, and every such subgroup is the kernel of exactly one. By Proposition 2.3, these subgroups correspond to the extensions of degree 2 inside k^{sep}, all Galois. So

$$k^\times/k^{\times 2} - \{1\} \xrightarrow{\delta} \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \{\pm 1\}) - \{1\} \to \{\text{quadratic extensions}\}$$

is a composition of bijections, and it maps d to the fixed field of $\mathrm{Gal}_{k(\sqrt{d})}$, which is $k(\sqrt{d})$. For $k = \mathbb{Q}$, the quadratic fields are the $\mathbb{Q}(\sqrt{d})$ with d a squarefree integer other than 1, each obtained once.

(5) Let L be an extension of k inside k^{sep}. Show that the restriction $H^1(k, \mu_m) \to H^1(L, \mu_m)$ corresponds, under δ, to the map $k^\times/k^{\times m} \to L^\times/L^{\times m}$ induced by inclusion, and that its kernel corresponds to $(k^\times \cap L^{\times m})/k^{\times m}$. Compute the kernel for $\mathbb{Q}(i)/\mathbb{Q}$ and $m = 2$, and compare it with Proposition 3.3.

Solution. **$\{\pm 1\}$, of order 2, as Proposition 3.3 predicts.** The field k^{sep} is also a separable closure of L, with $\mathrm{Gal}_L \subseteq \mathrm{Gal}_k$. For $a \in k^\times$ and $\alpha^m = a$, the restriction of $\sigma \mapsto \sigma(\alpha)/\alpha$ to Gal_L is the same formula, which is $\delta_L(a)$. So $\mathrm{Res} \circ \delta_k = \delta_L$ on $k^\times$, and since δ_L is injective, the kernel of Res is the image of the classes a with $a \in L^{\times m}$.
For $k = \mathbb{Q}$, $L = \mathbb{Q}(i)$ and $m = 2$: if $a \in \mathbb{Q}^\times$ equals $(x + iy)^2$ with $x, y \in \mathbb{Q}$, then $2xy = 0$, so $a = x^2$ or $a = -y^2$. Conversely $-1 = i^2$. So the kernel is $\{\pm 1\}\mathbb{Q}^{\times 2}/\mathbb{Q}^{\times 2}$, of order 2. By Proposition 3.3, it is $H^1(\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q}), \mu_2(\mathbb{Q}(i))) = \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \{\pm 1\})$ (Proposition 1.2), also of order 2.

4. Kummer theory

This section is [Sil09, Exercise 8.4], in the form of [Mil22, Theorem 5.30 and Remark 5.32], together with an example that shows why the roots of unity are needed. In it, $\mu_m(k^{\mathrm{sep}}) \subseteq k$, except in item (4).

Proposition 4.1 (Characters of a finite abelian group [Mil22, discussion before Theorem 5.30]). *If B is a finite abelian group of exponent dividing m, then $\mathrm{Hom}(B, \mu_m(k^{\mathrm{sep}}))$ has the same order as B.*

Problem 4.2 (Abelian extensions of exponent dividing m).

(1) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Show that δ is an isomorphism

$$\delta\colon k^\times/k^{\times m} \to \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}})).$$

For $a \in k^\times$ and $\alpha^m = a$, show that $\ker \delta(a) = \mathrm{Gal}_{k(\alpha)}$, and deduce that $k(\alpha)/k$ is Galois and cyclic, of degree the order of a in $k^\times/k^{\times m}$.

Solution. Now Gal_k acts trivially on $\mu_m(k^{\mathrm{sep}})$, so

$$H^1(k, \mu_m) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}}))$$

by Proposition 1.2, and Problem 3.4(1) gives the isomorphism. The homomorphism $\chi := \delta(a)$, $\sigma \mapsto \sigma(\alpha)/\alpha$, has kernel the σ fixing α, that is, $\mathrm{Gal}_{k(\alpha)}$. This kernel is normal, so $k(\alpha)/k$ is Galois with group $\mathrm{Gal}_k \,/\, \mathrm{Gal}_{k(\alpha)} \cong \chi(\mathrm{Gal}_k)$ (Proposition 2.3), a subgroup of the cyclic group $\mu_m(k^{\mathrm{sep}})$. A homomorphism into a cyclic group has order equal to the order of its image, and δ is an isomorphism, so $[k(\alpha) : k] = \#\chi(\mathrm{Gal}_k)$ is the order of a.

(2) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Let Δ be a subgroup of $k^\times$ containing $k^{\times m}$, with $\Delta/k^{\times m}$ finite, and $L := k(\Delta^{1/m})$ the extension of k inside k^{sep} generated by the m-th roots of the elements of Δ. Show that L/k is finite and Galois, and that

$$\langle\ ,\ \rangle\colon \mathrm{Gal}(L/k) \times \Delta/k^{\times m} \to \mu_m(k^{\mathrm{sep}}), \qquad \langle\sigma, a\rangle := \sigma(\alpha)/\alpha \quad (\alpha^m = a),$$

is a well defined bilinear pairing, and that it is perfect: the induced maps

$$\mathrm{Gal}(L/k) \to \mathrm{Hom}(\Delta/k^{\times m}, \mu_m(k^{\mathrm{sep}})), \qquad \Delta/k^{\times m} \to \mathrm{Hom}(\mathrm{Gal}(L/k), \mu_m(k^{\mathrm{sep}}))$$

are isomorphisms. Deduce that $\mathrm{Gal}(L/k)$ is abelian of exponent dividing m and that $[L : k] = (\Delta : k^{\times m})$.

Solution. Let $a_1, \dots, a_r$ represent generators of $\Delta/k^{\times m}$. Every $a \in \Delta$ is $a_1^{e_1} \cdots a_r^{e_r} c^m$ with $c \in k^\times$, so L is generated by the roots of the separable polynomials $x^m - a_i$. All the roots of $x^m - a_i$ are $\zeta\alpha_i$ with $\zeta \in \mu_m(k^{\mathrm{sep}}) \subseteq k$, so L is the splitting field of $\prod_i (x^m - a_i)$: finite and Galois.

The value $\sigma(\alpha)/\alpha$ lies in $\mu_m(k^{\mathrm{sep}})$, and does not depend on α, since σ fixes $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Replacing a by ac^m replaces α by αc, with the same value. So the pairing is well defined. It is multiplicative in a, as in Problem 3.4(2), and in σ, since $(\sigma\tau)(\alpha)/\alpha = \sigma(\tau(\alpha)/\alpha) \cdot \sigma(\alpha)/\alpha$ and $\tau(\alpha)/\alpha \in k$ is fixed by σ.

If $\langle\sigma, a\rangle = 1$ for every a, then σ fixes all the generators of L, so $\sigma = 1$. If $\langle\sigma, a\rangle = 1$ for every σ, then $\alpha \in L^{\mathrm{Gal}(L/k)} = k$, so $a \in k^{\times m}$. So both induced maps are injective. The first embeds $\mathrm{Gal}(L/k)$ in an abelian group of exponent dividing m, so $\mathrm{Gal}(L/k)$ is one too. By Proposition 4.1, the two injections give $[L : k] \leqslant (\Delta : k^{\times m}) \leqslant [L : k]$. So equality holds, and both injections are bijections.

(3) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Show that $\Delta \mapsto k(\Delta^{1/m})$ is a bijection from the subgroups Δ as in (2) to the finite abelian extensions of k inside k^{sep} of exponent dividing m, with inverse $L \mapsto \Delta_L := k^\times \cap L^{\times m}$.

SOLUTION. Let L be a finite abelian extension of exponent dividing m. By Problem 3.4(5) and Proposition 3.3, with $\mu_m(k^{\mathrm{sep}})^{\mathrm{Gal}_L} = \mu_m(L) = \mu_m(k^{\mathrm{sep}})$,

$$\Delta_L/k^{\times m} \cong \ker\big(H^1(k,\mu_m) \to H^1(L,\mu_m)\big) \cong H^1(\mathrm{Gal}(L/k),\mu_m(L)) = \mathrm{Hom}(\mathrm{Gal}(L/k),\mu_m(L)),$$

which has order $[L:k]$ by Proposition 4.1. Every element of Δ_L is an m-th power in L, and then all its m-th roots lie in L, as $\mu_m(k^{\mathrm{sep}}) \subseteq k$. So $L' := k(\Delta_L^{1/m}) \subseteq L$, and $[L':k] = (\Delta_L : k^{\times m}) = [L:k]$ by (2). Hence $L' = L$. Conversely, let $L = k(\Delta^{1/m})$. Then $\Delta \subseteq \Delta_L$, and both have index $[L:k]$ over $k^{\times m}$, by (2) and the computation above. So $\Delta = \Delta_L$.

Remark. For $k = \mathbb{Q}$ and $m = 2$: the finite abelian extensions of $\mathbb{Q}$ of exponent 2 are the $\mathbb{Q}(\sqrt{d_1},\dots,\sqrt{d_r})$, indexed by the finite subgroups of $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ [Mil22, Example 5.31].

(4) ($\star$) Let $k = \mathbb{Q}$, $m = 3$, $\zeta \in \overline{\mathbb{Q}}$ a primitive cube root of unity, and $F := \mathbb{Q}(\zeta)$. Here $\mu_3(\overline{\mathbb{Q}}) \not\subseteq \mathbb{Q}$.

(a) Show that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not Galois, although $2 \notin \mathbb{Q}^{\times 3}$. Which step of (1) fails?

SOLUTION. **Proposition 1.2 fails: $\mathrm{Gal}_{\mathbb{Q}}$ acts nontrivially on $\mu_3(\overline{\mathbb{Q}})$.** The real field $\mathbb{Q}(\sqrt[3]{2})$ does not contain the other roots $\zeta\sqrt[3]{2}$, $\zeta^2\sqrt[3]{2}$ of the irreducible polynomial x^3-2, so it is not normal. In (1), Proposition 1.2 no longer applies: complex conjugation inverts ζ. The set $\{\sigma : \sigma(\alpha) = \alpha\} = \mathrm{Gal}_{\mathbb{Q}(\alpha)}$, for $\alpha = \sqrt[3]{2}$, is still the set where the cocycle $\xi_\sigma = \sigma(\alpha)/\alpha$ is 1, and it is still a subgroup, but no longer the kernel of a homomorphism, and it is not normal.

(b) Show that $\mathbb{Q}^\times/\mathbb{Q}^{\times 3} \to F^\times/F^{\times 3}$ is injective.

SOLUTION. By Problem 3.4(5) and Proposition 3.3, the kernel is isomorphic to

$$H^1(\mathrm{Gal}(F/\mathbb{Q}),\mu_3(F)).$$

This group is killed by $\#\,\mathrm{Gal}(F/\mathbb{Q}) = 2$ (Problem 1.3(4)), and by 3, since the cocycles take values in $\mu_3(F)$. So it is trivial.

Remark. An elementary proof: if $a = \beta^3$ with $\beta \in F$, then $a^2 = N_{F/\mathbb{Q}}(a) = N_{F/\mathbb{Q}}(\beta)^3$, so $a = a^3/a^2 = (a/N_{F/\mathbb{Q}}(\beta))^3 \in \mathbb{Q}^{\times 3}$.

(c) Let $a \in \mathbb{Q}^\times - \mathbb{Q}^{\times 3}$, $\alpha^3 = a$, and $L := F(\alpha)$. Show that $L/\mathbb{Q}$ is Galois of degree 6, and that conjugation by any $\tau \in \mathrm{Gal}(L/\mathbb{Q})$ with $\tau(\zeta) = \zeta^{-1}$ acts on $\mathrm{Gal}(L/F) \cong \mu_3(F)$, the isomorphism of (1) over F, as inversion. Deduce that $\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$. Check with `Magma`:

```
R<x> := PolynomialRing(Rationals());
G := GaloisGroup(x^3 - 2);
#G, IsAbelian(G);                    // 6 false
F<zeta> := CyclotomicField(3);
S<y> := PolynomialRing(F);
#GaloisGroup(y^3 - 2);               // 3
```

SOLUTION. The field L is generated by the roots $\alpha, \zeta\alpha, \zeta^2\alpha$ of $x^3 - a$, so it is its splitting field, Galois over $\mathbb{Q}$. By (4)(b), a is not a cube in F, so $[L:F] = 3$ by (1) over F, and $[L:\mathbb{Q}] = 6$. Let $\gamma \in \mathrm{Gal}(L/F)$ and $\omega := \gamma(\alpha)/\alpha \in \mu_3(F)$. The element $\beta := \tau^{-1}\alpha$ is another cube root of a, so $\gamma(\beta)/\beta = \omega$ too, as γ fixes $\mu_3(F) \subseteq F$. Then

$$(\tau\gamma\tau^{-1})(\alpha) = \tau(\gamma(\beta)) = \tau(\omega\beta) = \tau(\omega)\,\alpha = \omega^{-1}\alpha.$$

So $\tau\gamma\tau^{-1}$ corresponds to ω^{-1}. Hence $\mathrm{Gal}(L/\mathbb{Q})$, of order 6, is nonabelian, so it is S_3. `Magma` agrees for $a = 2$: the group has order 6 over $\mathbb{Q}$, is not abelian, and has order 3 over F.

Remark. So $\mathrm{Gal}(L/F) \cong \mu_3(F)$ is an isomorphism of $\mathrm{Gal}(F/\mathbb{Q})$-modules, for the action by conjugation on the left and the natural action on the right, and it is the nontrivial action on $\mu_3(F)$ that makes $L/\mathbb{Q}$ nonabelian. In cohomological terms, (4)(b) says that Res: $H^1(\mathbb{Q}, \mu_3) \to H^1(F, \mu_3)$ is injective, and its image lies in the $\mathrm{Gal}(F/\mathbb{Q})$-invariants (compare [Sil09, Exercise B.3], for finite groups).

5. Computing $k^\times/k^{\times m}$

This section computes $k^\times/k^{\times m} \cong H^1(k, \mu_m)$ for finite fields, $\mathbb{Q}_p$ and $\mathbb{Q}$, and then cuts out a finite subgroup by conditions at the primes, as in [Poo02, §§9–10]. The last item is [Sil09, proof of Proposition VIII.1.6], with a `Magma` check.

Lemma 5.1 (Hensel's lemma [Sil09, Lemma IV.1.2])**.** *Let $F \in \mathbb{Z}_p[w]$ and $a \in \mathbb{Z}_p$ with $F(a) \in p^n\mathbb{Z}_p$ for some $n \geqslant 1$ and $F'(a) \in \mathbb{Z}_p^\times$. Then F has a root $b \in \mathbb{Z}_p$ with $b \equiv a \pmod{p^n}$.*

Definition 5.2 (Unramified classes [Poo02, §§4 and 9])**.** Let p be a prime. Embed $\overline{\mathbb{Q}}$ in an algebraic closure $\overline{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, so that restriction is an injection $\mathrm{Gal}_{\mathbb{Q}_p} := \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}_{\mathbb{Q}}$. Let ${\mathbb{Q}_p}^{\mathrm{unr}} \subseteq \overline{\mathbb{Q}}_p$ be the maximal unramified extension of $\mathbb{Q}_p$, the union of its finite unramified extensions. A finite extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}}_p$ is unramified if and only if it lies in ${\mathbb{Q}_p}^{\mathrm{unr}}$. The inertia group is $I_p := \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}}) \subseteq \mathrm{Gal}_{\mathbb{Q}_p}$. A class $\xi \in H^1(\mathbb{Q}, A)$, for a commutative algebraic group A over $\mathbb{Q}$, is unramified at p if its image in $H^1(I_p, A(\overline{\mathbb{Q}}_p))$ is trivial. For a finite set S of primes, $H^1(\mathbb{Q}, A; S)$ is the subgroup of the classes unramified at every prime $p \notin S$.

Proposition 5.3 (Unramified Kummer extensions [Sil09, proof of Proposition VIII.1.6])**.** *Let p be a prime with $p \nmid m$, $a \in {\mathbb{Q}_p}^\times$ and $\alpha \in \overline{\mathbb{Q}}_p$ with $\alpha^m = a$. Then $\mathbb{Q}_p(\alpha)/\mathbb{Q}_p$ is unramified if and only if $v_p(a) \equiv 0 \pmod{m}$.*

Problem 5.4 (Finite and local fields)**.**

(1) Let q be a power of a prime not dividing m. Show that $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ is cyclic of order $\gcd(m, q-1)$. If $m \mid q-1$, so that $\mu_m(\overline{\mathbb{F}}_q) \subseteq \mathbb{F}_q$, compare the subgroups of $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ with the extensions of $\mathbb{F}_q$ of degree dividing m, as Problem 4.2(3) predicts.

Solution. **Both are indexed by the divisors of m.** By Proposition 3.1, $\mathbb{F}_q^\times$ is cyclic of order $q-1$. In a cyclic group C of order n, written additively as $\mathbb{Z}/n\mathbb{Z}$, the subgroup mC is generated by m, of order $n/\gcd(m,n)$, so C/mC is cyclic of order $\gcd(m,n)$. If $m \mid q-1$, the group $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ is cyclic of order m, and its subgroups are indexed by the divisors d of m, with the subgroup of order d. On the other side, $\mathbb{F}_q$ has exactly one extension of each degree d inside an algebraic closure, $\mathbb{F}_{q^d}$, cyclic over $\mathbb{F}_q$, and it has exponent dividing m exactly when $d \mid m$. The bijection of Problem 4.2(3) matches the subgroup of order d with $\mathbb{F}_{q^d}$.

(2) Let p be odd and $u \in \mathbb{Z}_p^\times$ a unit whose reduction is not a square in $\mathbb{F}_p$. Show that ${\mathbb{Q}_p}^\times/{\mathbb{Q}_p}^{\times 2}$ has order 4, with representatives $1, u, p, up$. List the quadratic extensions of $\mathbb{Q}_p$, and say which one is unramified.

Solution. $\mathbb{Q}_p(\sqrt{u})$, $\mathbb{Q}_p(\sqrt{p})$, $\mathbb{Q}_p(\sqrt{up})$. **Only $\mathbb{Q}_p(\sqrt{u})$ is unramified.** If $v \in \mathbb{Z}_p^\times$ reduces to a square $\bar{c}^2$ in $\mathbb{F}_p$, with $c \in \mathbb{Z}_p^\times$, then $F(w) := w^2 - v$ has $F(c) \in p\mathbb{Z}_p$ and $F'(c) = 2c \in \mathbb{Z}_p^\times$, so v is a square by Lemma 5.1. Conversely, squares of units reduce to squares. So $\mathbb{Z}_p^\times/\mathbb{Z}_p^{\times 2} \cong \mathbb{F}_p^\times/\mathbb{F}_p^{\times 2}$, of order 2 by (1), with representatives $1, u$. As ${\mathbb{Q}_p}^\times = p^{\mathbb{Z}} \times \mathbb{Z}_p^\times$ and p^2 is a square, ${\mathbb{Q}_p}^\times/{\mathbb{Q}_p}^{\times 2}$ has order 4, with representatives $1, u, p, up$. By Problem 3.4(4), the quadratic extensions are

$\mathbb{Q}_p(\sqrt{d})$ for $d = u, p, up$. By Proposition 5.3, the unramified one is $\mathbb{Q}_p(\sqrt{u})$, as $v_p(u) = 0$ is even.

(3) Show that a unit $v \in \mathbb{Z}_2^\times$ is a square if and only if $v \equiv 1 \pmod 8$. Deduce that $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$ has order 8, with representatives $\pm 1, \pm 2, \pm 5, \pm 10$. `Magma` counts the quadratic extensions of $\mathbb{Q}_p$:

```
#AllExtensions(pAdicRing(3, 20), 2);    // 3
#AllExtensions(pAdicRing(5, 20), 2);    // 3
#AllExtensions(pAdicRing(2, 20), 2);    // 7
```

Check these against your answers.

SOLUTION. **They agree: 3 for odd p, and $8 - 1 = 7$ for $p = 2$.** An odd integer has square $\equiv 1 \pmod 8$, and the same holds in $\mathbb{Z}_2$, by continuity, so squares of units are $\equiv 1 \pmod 8$. Conversely, let $v \equiv 1 \pmod 8$ and $c := (v-1)/4 \in 2\mathbb{Z}_2$. Then $F(w) = w^2 + w - c$ has $F(0) = -c \in 2\mathbb{Z}_2$ and $F'(0) = 1$, so it has a root $b \in \mathbb{Z}_2$ by Lemma 5.1, and $(2b+1)^2 = 4(b^2+b) + 1 = 4c + 1 = v$. Hence $\mathbb{Z}_2^\times/\mathbb{Z}_2^{\times 2} \cong (\mathbb{Z}/8\mathbb{Z})^\times = \{\pm 1, \pm 5\}$, and $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$, with $\mathbb{Q}_2^\times = 2^{\mathbb{Z}} \times \mathbb{Z}_2^\times$, has the eight representatives $\pm 1, \pm 5, \pm 2, \pm 10$. By Problem 3.4(4), the quadratic extensions of $\mathbb{Q}_p$ correspond to the nontrivial classes: 3 for odd p, by (2), and 7 for $p = 2$.

Remark. Proposition 5.3 does not apply to $p = 2$ and $m = 2$. Of the seven quadratic extensions of $\mathbb{Q}_2$, only $\mathbb{Q}_2(\sqrt{5}) = \mathbb{Q}_2(\sqrt{-3})$ is unramified.

Definition 5.5 (The group $k(S, m)$ [Sil09, Theorem X.1.1(c)])**.** Let k be a number field, S a finite set of places of k containing the archimedean ones, and

$$k(S, m) := \{a \in k^\times/k^{\times m} : v(a) \equiv 0 \pmod m \text{ for all } v \notin S\},$$

where v runs over the normalized valuations of k. Let $\mathcal{O}_S$ be the ring of S-integers, the $a \in k$ with $v(a) \geqslant 0$ for all $v \notin S$, and $\mathrm{Cl}(\mathcal{O}_S)$ its ideal class group. For $k = \mathbb{Q}$ we list only the primes in S, and in [Poo02, §10], $\mathbb{Q}(S, 2)$ is written $\langle -1, S\rangle$.

Proposition 5.6 (Finiteness [Sil09, proof of Proposition VIII.1.6])**.** *The group* $\mathrm{Cl}(\mathcal{O}_S)$ *is finite, and* $\mathcal{O}_S^\times$ *is finitely generated (Dirichlet's S-unit theorem).*

Problem 5.7 (Classes unramified outside S)**.**

(1) Show that $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ is an $\mathbb{F}_2$-vector space with basis the classes of -1 and of the primes. In particular, it is infinite.

SOLUTION. Every $a \in \mathbb{Q}^\times$ is $\pm\prod_p p^{v_p(a)}$, uniquely. Modulo squares only the sign and the parities of the $v_p(a)$ matter, so $a \mapsto (\operatorname{sign} a, (v_p(a) \bmod 2)_p)$ is an isomorphism from $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ to $\{\pm 1\} \times \bigoplus_p \mathbb{Z}/2\mathbb{Z}$, where the sum is over the primes.

(2) Let S be a finite set of primes. Show that $\mathbb{Q}(S, 2)$ is the subgroup generated by -1 and the primes in S, of order $2^{\#S+1}$.

SOLUTION. In the coordinates of (1), $\mathbb{Q}(S, 2)$ is the set of classes whose coordinates at the primes outside S vanish, which is the span of -1 and the primes in S.

(3) Let p be an odd prime and $a \in \mathbb{Q}^\times$. Show that $\delta(a) \in H^1(\mathbb{Q}, \mu_2)$ is unramified at p if and only if $v_p(a)$ is even. Deduce that $\mathbb{Q}(S, 2) = \delta^{-1}H^1(\mathbb{Q}, \mu_2; S)$ for every finite set S of primes containing 2.

Solution. The action of I_p on $\mu_2(\overline{\mathbb{Q}}_p) = \{\pm 1\}$ is trivial, so $H^1(I_p, \mu_2(\overline{\mathbb{Q}}_p)) = \mathrm{Hom}_{\mathrm{conts}}(I_p, \{\pm 1\})$ (Proposition 1.2), and the restriction of $\delta(a)$ is the homomorphism $\sigma \mapsto \sigma(\sqrt{a})/\sqrt{a}$. It is trivial if and only if $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}})$ fixes $\sqrt{a}$, that is, if and only if $\sqrt{a} \in {\mathbb{Q}_p}^{\mathrm{unr}}$ (Proposition 2.3 over $\mathbb{Q}_p$), that is, by Definition 5.2, if and only if $\mathbb{Q}_p(\sqrt{a})/\mathbb{Q}_p$ is unramified. By Proposition 5.3, with $p \nmid 2$, this happens exactly when $v_p(a)$ is even. So for $S \ni 2$, $\delta(a)$ is unramified at every $p \notin S$ exactly when $v_p(a)$ is even for every $p \notin S$, that is, when $a \in \mathbb{Q}(S, 2)$.

Remark. This is the computation behind 2-descent: [Poo02, §10] identifies $H^1(\mathbb{Q}, E[2]; S)$ with $\langle -1, S\rangle^2$ when $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$.

(4) ($\star$) Let k be a number field and S as in Definition 5.5. Construct an exact sequence

$$1 \to \mathcal{O}_S^\times/\mathcal{O}_S^{\times m} \to k(S, m) \to \mathrm{Cl}(\mathcal{O}_S)[m] \to 0,$$

and deduce that $k(S, m)$ is finite. For $k = \mathbb{Q}(\sqrt{-5})$, $m = 2$ and S the archimedean place, show that $k(S, 2) = \langle -1, 2\rangle$, of order 4, and check with `Magma`:

```
K<s> := QuadraticField(-5);
OK := Integers(K);
#ClassGroup(OK);                        // 2
G, m := pSelmerGroup(2, {PowerIdeal(OK)|});
#G;                                     // 4
[K | G.i @@ m : i in [1..Ngens(G)]];  // [ -1, 2 ]
```

Solution. **$a \mapsto [\mathfrak{a}]$, where $a\mathcal{O}_S = \mathfrak{a}^m$.** If a represents an element of $k(S, m)$, its fractional ideal $a\mathcal{O}_S$ has all exponents divisible by m, so it is $\mathfrak{a}^m$ for a unique fractional ideal $\mathfrak{a}$. Replacing a by ac^m replaces $\mathfrak{a}$ by $c\mathfrak{a}$, so $a \mapsto [\mathfrak{a}]$ is a well defined homomorphism to $\mathrm{Cl}(\mathcal{O}_S)$, with values in $\mathrm{Cl}(\mathcal{O}_S)[m]$, as $\mathfrak{a}^m$ is principal. It is surjective: if $\mathfrak{a}^m = a\mathcal{O}_S$, then a represents an element of $k(S, m)$ mapping to $[\mathfrak{a}]$. Its kernel: if $\mathfrak{a} = c\mathcal{O}_S$, then $a/c^m \in \mathcal{O}_S^\times$, so the kernel is the image of $\mathcal{O}_S^\times$. Finally, $u \in \mathcal{O}_S^\times$ maps to 1 in $k^\times/k^{\times m}$ if $u = c^m$ with $c \in k^\times$, and then $c \in \mathcal{O}_S^\times$, since $mv(c) = v(u) = 0$ for $v \notin S$. So $\mathcal{O}_S^\times/\mathcal{O}_S^{\times m}$ injects. Both ends are finite by Proposition 5.6, so $k(S, m)$ is finite.

For $k = \mathbb{Q}(\sqrt{-5})$ and S archimedean, $\mathcal{O}_S = \mathbb{Z}[\sqrt{-5}]$, with unit group $\{\pm 1\}$, so $\mathcal{O}_S^\times/\mathcal{O}_S^{\times 2}$ is generated by -1. The class group has order 2 (`Magma`), generated by $\mathfrak{p} := (2, 1 + \sqrt{-5})$, which is not principal, as no element of $\mathbb{Z}[\sqrt{-5}]$ has norm 2. And $\mathfrak{p}^2 = (2)$. So 2 represents an element of $k(S, 2)$ mapping to $[\mathfrak{p}]$, and $k(S, 2) = \langle -1, 2\rangle$, of order 4, as `Magma` says.

Remark. In the notation of Definition 5.2, extended to k, $k(S, m) = H^1(k, \mu_m; S)$ when S contains the places above m, and the sequence is the Kummer sequence for the étale cohomology of $\mathrm{Spec}\,\mathcal{O}_S$, with $\mathrm{Cl}(\mathcal{O}_S) = \mathrm{Pic}\,\mathcal{O}_S$ in place of $H^1(k, \mathbb{G}_m) = 1$. Compare the analogy between $\mathrm{Cl}(k)$ and the Shafarevich–Tate group in [Poo02, §7].

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